

1) Setup Jenkins CICD pipeline using freestyle job using Docker containers using below code.

Source Code: <https://github.com/betawins/hiring-app.git>

Stages:

- **Git Clone**
- **SonarQube Integration**
- **Maven Compilation**
- **Nexus Artifactory**
- **Slack Notification**
- **Deploy On tomcat**

Implementation Steps

Jenkins Freestyle Job Creation

1. Log in to the Jenkins server.
2. Open the Jenkins Dashboard.
3. Click **New Item**.
4. Enter the job name:
Hiring-App-Freestyle-Docker
5. Select **Freestyle project**.
6. Click **OK**.

Configure Git Clone

7. Open **Source Code Management**.
8. Select **Git**.
9. Enter the repository URL:

<https://github.com/betawins/hiring-app.git>

10. Set the branch:

*/main

11. Since the repository is public, Git credentials are not required.
12. Click **Apply**.

Configure Docker

13. Ensure Docker is installed on the Jenkins server.

14. Verify Docker:

```
docker --version
```

15. Verify that the Jenkins user can access Docker:

```
sudo -u jenkins docker ps
```

16. Pull the Maven Docker image:

```
docker pull maven:3.9.9-eclipse-temurin-21
```

17. Verify the Maven Docker image:

```
docker images
```

Git Clone Verification

18. Go to **Build Steps**.

19. Select **Execute shell**.

20. Add the following commands:

```
echo "==== Git Clone Verification ===="
```

```
echo "Workspace:"
```

```
pwd
```

```
echo "==== Files in Workspace ===="
```

```
ls -la
```

```
echo "==== Git Commit ===="
```

```
git log -1 --oneline
```

```
echo "==== pom.xml ===="
```

```
ls -l pom.xml
```

21. Save the configuration.

SonarQube Configuration

22. Go to **Manage Jenkins → System**.

23. Locate **SonarQube servers**.

24. Add the SonarQube server.

25. Set the SonarQube server name:

SonarQube

26. Set the SonarQube server URL:

http://172.31.12.251:9000/

27. Select the SonarQube authentication credential.

28. Click **Apply**.

SonarQube Analysis

29. Return to the Jenkins job configuration.

30. Add another **Build Step**.

31. Select **Execute SonarQube Scanner**.

32. Configure the analysis properties:

sonar.projectKey=hiring-app

sonar.projectName=Hiring-App

sonar.sources=src

33. Save the configuration.

Maven Build Using Docker

34. Add another **Build Step**.

35. Select **Execute shell**.

36. Add the Maven Docker build commands:

```
echo "===== Maven Version ====="
```

```
docker run --rm \
```

```
-v "$WORKSPACE:/workspace" \
```

```
-w /workspace \
```

```
maven:3.9.9-eclipse-temurin-21 \
```

```
mvn --version
```

```
echo "===== Maven Build ====="
```

```
docker run --rm \
-v "$WORKSPACE:/workspace" \
-w /workspace \
maven:3.9.9-eclipse-temurin-21 \
mvn clean install
```

```
echo "==== Build Artifacts ====="
```

```
ls -lh target/
```

37. Save the configuration.

Verify Maven Artifact

38. Verify that Maven generates the WAR file:

```
target/hiring.war
```

39. Verify the Maven build result in the Jenkins console:

```
BUILD SUCCESS
```

Nexus Repository Configuration

40. Open the Nexus Repository server.

41. Verify that the Maven repository is available.

42. Use the repository:

```
maven_releases
```

43. Configure the Nexus credentials in **Jenkins → Credentials**.

44. Install the **Nexus Artifact Uploader** plugin if it is not already installed.

Configure Nexus Artifact Upload

45. Open the Jenkins job configuration.

46. Go to **Post-build Actions**.

47. Select **Nexus Artifact Uploader**.

48. Set the Nexus version:

NEXUS3

49. Set the protocol:

HTTP

50. Set the Nexus URL:

172.31.47.243:8081

51. Select the Nexus repository credentials.

52. Configure the artifact details:

GroupId: in.javahome

Artifact ID: hiring

Version: 0.1

Repository: maven_releases

Classifier: empty

Type: war

File: target/hiring.war

53. Save the configuration.

Slack Notification Configuration

54. Ensure the **Slack Notification** plugin is installed.

55. Configure the Slack integration in Jenkins.

56. Add the Slack authentication credential.

57. Configure the required Slack notification settings.

58. Configure the Jenkins job to send a successful build notification.

59. Save the configuration.

Tomcat Server Configuration

60. Install Tomcat under:

/opt/apache-tomcat-9.0.122

61. Start Tomcat:

cd /opt/apache-tomcat-9.0.122/bin

./startup.sh

62. Verify that Tomcat is listening on port 8080:

```
ss -lntp | grep 8080
```

63. Configure the Tomcat Manager user in:

```
/opt/apache-tomcat-9.0.122/conf/tomcat-users.xml
```

64. Configure the required Tomcat Manager roles:

```
manager-gui
```

```
manager-script
```

65. Configure the Jenkins deployment user:

```
jenkins
```

66. Configure the Tomcat Manager remote-access restriction to allow the Jenkins server to access the Manager application.

Verify Tomcat Manager

67. Test Tomcat Manager authentication:

```
curl -4 -u jenkins http://127.0.0.1:8080/manager/text/list
```

68. Verify that the response contains:

```
OK - Listed applications for virtual host localhost
```

Jenkins Tomcat Deployment Configuration

69. Ensure the **Deploy to container** Jenkins plugin is installed.

70. Open the Jenkins job configuration.

71. Go to **Post-build Actions**.

72. Select **Deploy war/ear to a container**.

73. Set the WAR/EAR file:

```
target/hiring.war
```

74. Select the container:

```
Tomcat 9.x
```

75. Add the Tomcat Jenkins credentials.

76. Set the Tomcat server URL:

```
http://172.31.33.74:8080
```

77. Click **Apply**.

78. Click **Save**.

Execute the Jenkins Job

79. Open the Jenkins job:

Hiring-App-Freestyle-Docker

80. Click **Build Now**.

81. Open the generated build.

82. Select **Console Output**.

83. Verify Git checkout is successful.

84. Verify SonarQube analysis is successful.

85. Verify Maven compilation and packaging are successful.

86. Verify target/hiring.war is generated.

87. Verify the WAR artifact is uploaded to Nexus.

88. Verify Tomcat deployment is successful.

89. Verify Slack notification is sent.

90. Verify the final Jenkins result:

Finished: SUCCESS

Verify Application Deployment

91. Open the Tomcat application URL:

<http://<TOMCAT-PUBLIC-IP>:8080/hiring/>

92. Verify that the Hiring application is accessible.

Jenkins / Hiring-App-Freestyle-Doc... / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/hiring-app.git

Credentials ?

- none - + Add

Advanced v

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Save Apply

Jenkins / Hiring-App-Freestyle-Doc... / #1

Status

Changes

Console Output

Delete build '#1'

Git Build Data

Edit build information

Console

Download Copy View as plain text

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Hiring-App-Freestyle-Docker
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/betawins/hiring-app.git
> /usr/bin/git init /var/lib/jenkins/workspace/Hiring-App-Freestyle-Docker # timeout=10
Fetching upstream changes from https://github.com/betawins/hiring-app.git
> /usr/bin/git --version # timeout=10
> git --version # 'git version 2.50.1'
> /usr/bin/git fetch --tags --force --progress -- https://github.com/betawins/hiring-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> /usr/bin/git config remote.origin.url https://github.com/betawins/hiring-app.git # timeout=10
> /usr/bin/git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> /usr/bin/git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 9405b107fdffdf68e69fe953eaf9669784f9d6a (refs/remotes/origin/main)
> /usr/bin/git config core.sparsecheckout # timeout=10
> /usr/bin/git checkout -f 9405b107fdffdf68e69fe953eaf9669784f9d6a # timeout=10
Commit message: "Update README.md"
First time build. skipping changelog.
[Hiring-App-Freestyle-Docker] $ /bin/sh -xe /tmp/jenkins4037758124452946561.sh
+ echo '==== Git Clone Verification ====='
===== Git Clone Verification =====
+ echo Workspace:
```

← → ↺ Not secure 100.54.71.26:8080/job/Hiring-App-Freestyle-Docker/1/console

Jenkins / Hiring-App-Freestyle-Doc... / #1

```
+ echo Workspace:
Workspace:
+ pwd
/var/lib/jenkins/workspace/Hiring-App-Freestyle-Docker
+ echo '==== Files in Workspace ====='
===== Files in Workspace =====
+ ls -la
total 28
drwxr-xr-x. 4 jenkins jenkins 171 Oct 7 10:33 .
drwxr-xr-x. 5 jenkins jenkins 107 Oct 7 10:33 ..
drwxr-xr-x. 7 jenkins jenkins 146 Oct 7 10:33 .git
-rw-r--r--. 1 jenkins jenkins 8 Oct 7 10:33 .gitignore
-rw-r--r--. 1 jenkins jenkins 476 Oct 7 10:33 Dockerfile
-rw-r--r--. 1 jenkins jenkins 1757 Oct 7 10:33 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins 88 Oct 7 10:33 README.md
-rw-r--r--. 1 jenkins jenkins 408 Oct 7 10:33 Untitled Diagram.drawio
-rw-r--r--. 1 jenkins jenkins 1801 Oct 7 10:33 jenkinsfile-cicd
-rw-r--r--. 1 jenkins jenkins 890 Oct 7 10:33 pom.xml
drwxr-xr-x. 3 jenkins jenkins 18 Oct 7 10:33 src
+ echo '==== Git Commit ====='
===== Git Commit =====
+ git log -1 --oneline
9405b10 Update README.md
+ echo '==== pom.xml ====='
===== pom.xml =====
+ ls -l pom.xml
-rw-r--r--. 1 jenkins jenkins 890 Oct 7 10:33 pom.xml
Finished: SUCCESS
```

← → ↻ Not secure 100.54.71.26:8080/job/Hiring-App-Freestyle-Docker/2/console ☆ ⓘ

Jenkins / Hiring-App-Freestyle-Doc... / #2

```
Progress (1): 147/193 kB
Progress (1): 164/193 kB
Progress (1): 180/193 kB
Progress (1): 193 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 5.2 MB/s)
[INFO] Installing /workspace/pom.xml to /root/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.pom
[INFO] Installing /workspace/target/hiring.war to /root/.m2/repository/in/javahome/hiring/0.1/hiring-0.1.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.166 s
[INFO] Finished at: 2026-10-07T10:42:13Z
[INFO] -----
+ echo '==== Build Artifacts ====='
===== Build Artifacts =====
+ ls -lh target/
total 4.0K
drwxr-xr-x. 4 root root 54 Oct 7 10:42 hiring
-rw-r--r--. 1 root root 1.7K Oct 7 10:42 hiring.war
drwxr-xr-x. 2 root root 28 Oct 7 10:42 maven-archiver
Finished: SUCCESS
```

← → ↻ Not secure 100.54.71.26:8080/job/Hiring-App-Freestyle-Docker/3/console ☆ ⓘ

Jenkins / Hiring-App-Freestyle-Doc... / #3

```
10:53:43.861 INFO 5 source files to be analyzed for the text and secrets analysis
10:53:43.873 INFO 5/5 source files have been analyzed for the text and secrets analysis
10:53:43.874 INFO Sensor TextAndSecretsSensor [text] (done) | time=502ms
10:53:43.877 INFO ----- Run sensors on project
10:53:44.135 INFO Sensor IaC Project Sensor [iac]
10:53:44.145 INFO Sensor IaC Project Sensor [iac] (done) | time=3ms
10:53:44.145 INFO Sensor JavaProjectSensor [java]
10:53:44.146 INFO Sensor JavaProjectSensor [java] (done) | time=1ms
10:53:44.146 INFO Sensor Zero Coverage Sensor
10:53:44.146 INFO Sensor Zero Coverage Sensor (done) | time=0ms
10:53:44.146 INFO ----- Gather SCA dependencies on project
10:53:44.146 INFO Dependency analysis skipped
10:53:44.153 INFO CPD Executor 1 file had no CPD blocks
10:53:44.154 INFO CPD Executor Calculating CPD for 0 files
10:53:44.154 INFO CPD Executor CPD calculation finished (done) | time=0ms
10:53:44.158 INFO SCM revision ID '9405b107dfdf68e69fe953eaf9669784f9dea'
10:53:44.286 INFO Analysis report generated in 120ms, dir size=295.7 kB
10:53:44.318 INFO Analysis report compressed in 27ms, zip size=34.7 kB
10:53:44.358 INFO Analysis report uploaded in 42ms
10:53:44.365 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
10:53:44.365 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
10:53:44.365 INFO More about the report processing at http://172.31.12.251:9000/api/ce/task?id=ddae0da8-d923-4b17-8f14-248cbd7aede8
10:53:44.366 INFO Analysis total time: 5.463 s
10:53:44.372 INFO SonarScanner Engine completed successfully
10:53:44.381 INFO EXECUTION SUCCESS
10:53:44.382 INFO Total time: 10.318s
Finished: SUCCESS
```

← → ↻ Not secure 184.195.163.53:9000/projects ☆ ⓘ

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

The way in which security, reliability, and maintainability counts and ratings are calculated has changed. [Learn more in SonarQube documentation](#)

SonarQube community Projects Issues Rules Quality profiles Quality gates Administration More

My Favorites All

Filters

Quality gate

Passed 2

Failed 0

Security

A ≥ 0 info issues 0

B ≥ 1 low issue 1

C ≥ 1 medium issue 1

D ≥ 1 high issue 0

E ≥ 1 blocker issue 0

Reliability

A ≥ 0 info issues 0

B ≥ 1 low issue 1

C ≥ 1 medium issue 1

D ≥ 1 high issue 0

E ≥ 1 blocker issue 0

Search projects Perspective Overall Status Sort by Name 2 project(s)

Hiring-App Public Passed

Last analysis: 36 seconds ago · 45 Lines of Code · XML, Docker, ...

2 Security 3 Reliability 0 Maintainability 0 Hotspots Reviewed 0 Coverage 0.0% Duplications

VProfile-1 Public Passed

Last analysis: 1 day ago · 1.8k Lines of Code · JSP, Java, ...

9 Security 40 Reliability 51 Maintainability 0 Hotspots Reviewed 0 Coverage 0.0% Duplications 2.6%

2 of 2 shown

```
aws
Search [Alt+S]
United States (N. Virginia) Poojith R (956035085469) Poojith R

Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-47-243 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-10-07 11:02:26 UTC; 35s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 28031 (dockerd)
      Tasks: 9
     Memory: 29.0M
        CPU: 353ms
    CGroup: /system.slice/docker.service
            └─28031 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Oct 07 11:02:26 ip-172-31-47-243.ec2.internal systemd[1]: Starting docker.service - Docker Application Container Engine...
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal dockerd[28031]: time="2026-10-07T11:02:26.198694258Z" level=info msg="Starting up"
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal dockerd[28031]: time="2026-10-07T11:02:26.265837032Z" level=info msg="Loading containe
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal dockerd[28031]: time="2026-10-07T11:02:26.718338805Z" level=info msg="Loading containe
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal dockerd[28031]: time="2026-10-07T11:02:26.755158180Z" level=info msg="Docker daemon" c
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal dockerd[28031]: time="2026-10-07T11:02:26.755338536Z" level=info msg="Daemon has compl
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal dockerd[28031]: time="2026-10-07T11:02:26.784774605Z" level=info msg="API listen on /r
Oct 07 11:02:26 ip-172-31-47-243.ec2.internal systemd[1]: Started docker.service - Docker Application Container Engine.

[ec2-user@ip-172-31-47-243 ~]$
```

i-0127782299e58311b (Nexus Server)

PublicIPs: 50.16.38.159 PrivateIPs: 172.31.47.243

Not secure 54.196.80.250:8081/admin/repository/repositories:maven_releases

sonatype nexus repository Community Edition

Search components or CVEs...

Settings

Repository

- Repositories
- Blob Stores
- Data Store
- Proprietary Repositories
- Content Selectors
- Cleanup Policies
- Routing Rules

Security

Support

System

IQ Server

Repositories / maven_releases

Delete repository Rebuild index

Settings

Name: maven_releases

Format: maven2

Type: hosted

URL: http://54.196.80.250:8081/repository/maven_releases/

Online: ☒ If checked, the repository accepts incoming requests

Maven 2

Version policy: Release

Layout policy:

Validate that all paths are maven artifact or metadata paths

Strict

Content Disposition:

Add Content-Disposition header as 'Attachment' to disable some content from being inline in a browser.

Attachment

Storage

Blob store: default

Jenkins / Manage Jenkins / Plugins

Plugins

- Updates 3
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Nexus Artifact Uploader ☒ Success

Loading plugin extensions ☒ Success

→ [Go back to the top page](#)
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

← → ↻ Not secure 100.54.71.26:8080/job/Hiring-App-Freestyle-Docker/4/console

Jenkins / Hiring-App-Freestyle-Doc... / #4

```
11:57:18.242 INFO Analysis report generated in 116ms, dir size=295.7 kB
11:57:18.260 INFO Analysis report compressed in 18ms, zip size=34.7 kB
11:57:18.286 INFO Analysis report uploaded in 27ms
11:57:18.291 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
11:57:18.291 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
11:57:18.295 INFO More about the report processing at http://172.31.12.251:9000/api/ce/task?id=4f759a53-f73a-45ae-a614-bbbf37c08968
11:57:18.300 INFO Analysis total time: 4.007 s
11:57:18.317 INFO SonarScanner Engine completed successfully
11:57:18.336 INFO EXECUTION SUCCESS
11:57:18.336 INFO Total time: 7.486s
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploading: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 6.0 kB/s)
Uploading artifact hiring.war completed.
Finished: SUCCESS
```

Jenkins 2.580.1

← → ↻ Not secure 54.196.80.250:8081/#browse/browse:maven_releases

sonatype nexus repository Community Edition

Search components or CVEs...

Browse / maven_releases

Upload component HTML View

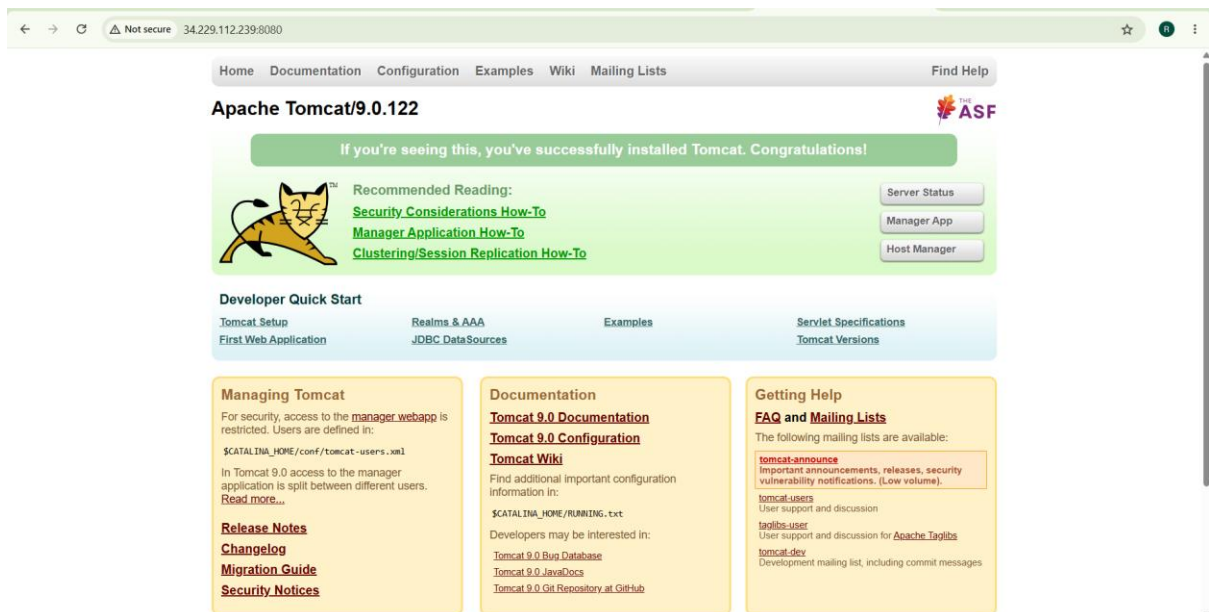
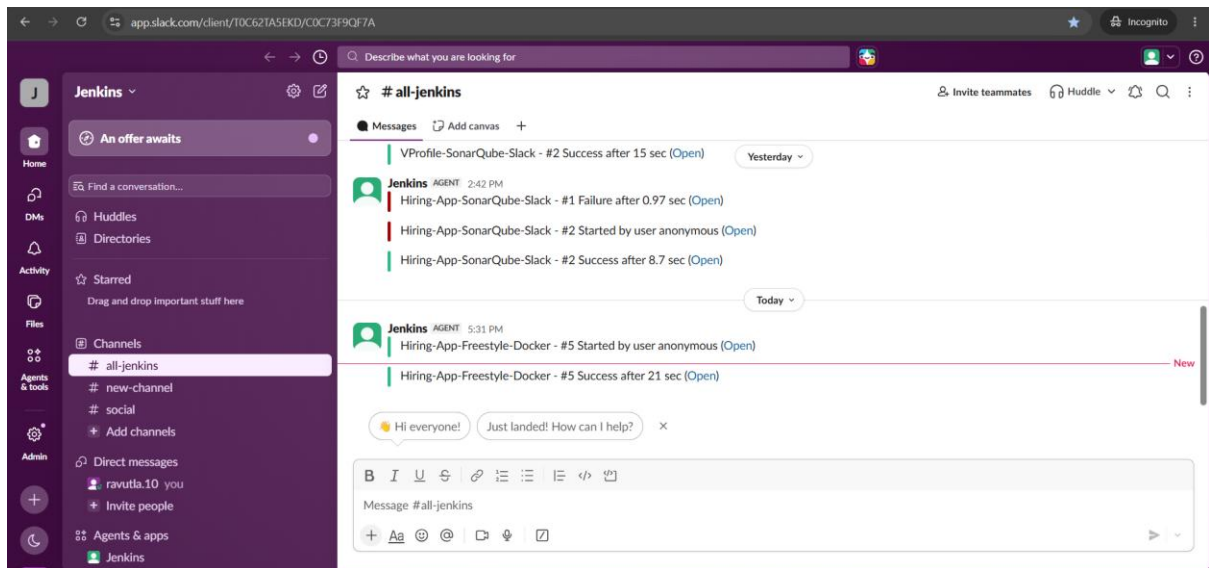
Advanced search...

- in
 - javahome
 - hiring
 - 0.1
 - hiring-0.1.war
 - hiring-0.1.war.md5
 - hiring-0.1.war.sha1

← → ↻ Not secure 100.54.71.26:8080/job/Hiring-App-Freestyle-Docker/5/console

Jenkins / Hiring-App-Freestyle-Doc... / #5

```
12:02:08.037 INFO Analysis report uploaded in 26ms
12:02:08.043 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=hiring-app
12:02:08.046 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
12:02:08.046 INFO More about the report processing at http://172.31.12.251:9000/api/ce/task?id=b703e6af-0e38-431b-a348-12b7733a956f
12:02:08.059 INFO Analysis total time: 4.090 s
12:02:08.064 INFO SonarScanner Engine completed successfully
12:02:08.074 INFO EXECUTION SUCCESS
12:02:08.075 INFO Total time: 7.683s
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploading: http://172.31.47.243:8081/repository/maven\_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 24 kB/s)
Uploading artifact hiring.war completed.
[Slack Notifications] found #4 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
```



The screenshot shows the Jenkins 'Manage Jenkins' page with the 'Plugins' tab selected. On the left, a sidebar lists 'Updates', 'Available plugins', 'Installed plugins', 'Advanced settings', and 'Download progress' (which is highlighted). The main area is titled 'Download progress' and shows a 'Preparation' step with three sub-items: 'Checking internet connectivity', 'Checking update center connectivity', and 'Success'. Below this, a table lists four items: 'Nexus Artifact Uploader', 'Loading plugin extensions', 'Deploy to container', and 'Loading plugin extensions', all with a green checkmark and 'Success' status. At the bottom, there are two links: 'Go back to the top page (you can start using the installed plugins right away)' and a checkbox for 'Restart Jenkins when installation is complete and no jobs are running'.

The screenshot shows the Jenkins console output for a build. The output text is as follows:
12:38:25.739 INFO SonarScanner Engine completed successfully
12:38:25.748 INFO EXECUTION SUCCESS
12:38:25.748 INFO Total time: 7.124s
Uploading artifact hiring.war started....
GroupId: in.javahome
ArtifactId: hiring
Classifier:
Type: war
Version: 0.1
File: hiring.war
Repository:maven_releases
Uploading: http://172.31.47.243:8081/repository/maven_releases/in/javahome/hiring/0.1/hiring-0.1.war
100 % completed (1.7 kB / 1.7 kB).
Uploaded: http://172.31.47.243:8081/repository/maven_releases/in/javahome/hiring/0.1/hiring-0.1.war (1.7 kB at 28 kB/s)
Uploading artifact hiring.war completed.
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/Hiring-App-Freestyle-Docker/target/hiring.war to container Tomcat 9.x Remote with context null
[/var/lib/jenkins/workspace/Hiring-App-Freestyle-Docker/target/hiring.war] is not deployed. Doing a fresh deployment.
Deploying [/var/lib/jenkins/workspace/Hiring-App-Freestyle-Docker/target/hiring.war]
[Slack Notifications] found #5 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS

The screenshot shows a web browser window with the address bar displaying '34.229.112.239:8080/hiring/'. The page content is 'Welcome to Techie Horizon'.

2) Setup a Jenkins CI/CD pipeline using Declarative pipeline using feature-1.1 branch.

Source Code:

https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1
stages:

- Git Clone
- SonarQube Integration

- **Maven Compilation**
- **Nexus Artifactory**
- **Slack Notification**
- **Deploy On tomcat**

Implementation Steps

1. Open **Jenkins Dashboard**.
2. Click **New Item**.
3. Enter:

CustomerApp-Declarative-Pipeline

4. Select **Pipeline**.
5. Click **OK**.
6. Scroll to the **Pipeline** section.
7. Select **Pipeline script**.
8. Use a **Declarative Pipeline** structure.

Pipeline Structure

pipeline

├— Git Clone

├— SonarQube Integration

├— Maven Compilation

├— Nexus Artifactory

├— Slack Notification

└— Deploy On Tomcat

9. Use the following Declarative Pipeline script:

```
pipeline {
```

```
    agent any
```

```
    stages {
```

```

// =====

// 1. GIT CLONE

// =====

stage('Git Clone') {
    steps {
        echo '==== Git Clone ====='

        git branch: 'feature-1.1',
            url: 'https://github.com/betawins/sabear_simplecutomerapp.git'

        echo '==== Repository Files ====='
        sh 'ls -la'

        echo '==== Git Commit ====='
        sh 'git log -1 --oneline'
    }
}

// =====

// 2. SONARQUBE INTEGRATION

// =====

stage('SonarQube Integration') {
    steps {
        echo '==== SonarQube Analysis ====='

        withSonarQubeEnv('SonarQube') {
            sh ""

```

```

        docker run --rm \
            -v "$WORKSPACE:/workspace" \
            -w /workspace \
            -e SONAR_HOST_URL \
            -e SONAR_AUTH_TOKEN \
            maven:3.9.9-eclipse-temurin-21 \
            mvn org.sonarsource.scanner.maven:sonar-maven-plugin:sonar \
            -Dsonar.projectKey=customer-app \
            -Dsonar.projectName=Customer-App \
            -Dsonar.host.url="$SONAR_HOST_URL" \
            -Dsonar.token="$SONAR_AUTH_TOKEN"
    ""
}
}
}

// =====
// 3. MAVEN COMPILATION
// =====

stage('Maven Compilation') {
    steps {
        echo '==== Maven Compilation ====='

        sh ""

        docker run --rm \
            -v "$WORKSPACE:/workspace" \
            -w /workspace \
            maven:3.9.9-eclipse-temurin-21 \

```

```

        mvn -DBUILD_NUMBER="$BUILD_NUMBER" clean package
    ""

    echo '==== Build Artifacts ====='

    sh 'ls -lh target/'

}

}

// =====

// 4. NEXUS ARTIFACT UPLOAD

// =====

stage('Nexus Artifactory') {
    steps {
        echo '==== Nexus Artifact Upload ====='

        nexusArtifactUploader(
            nexusVersion: 'nexus3',
            protocol: 'http',
            nexusUrl: '172.31.47.243:8081',
            groupId: 'com.javatpoint',
            version: "${BUILD_NUMBER}",
            repository: 'maven_releases',
            credentialsId: 'nexus-admin',
            artifacts: [
                [
                    artifactId: 'SimpleCustomerApp',
                    classifier: '',
                    file: "target/SimpleCustomerApp-${BUILD_NUMBER}-SNAPSHOT.war",

```

```

        type: 'war'
    ]
]
)
}
}

// =====

// 5. SLACK NOTIFICATION

// =====

stage('Slack Notification') {
    steps {
        echo '==== Slack Notification ====='

        slackSend(
            color: 'good',
            message: "CustomerApp Declarative Pipeline - Build #${BUILD_NUMBER} -
Nexus upload completed successfully."
        )
    }
}

// =====

// 6. DEPLOY ON TOMCAT

// =====

stage('Deploy On Tomcat') {
    steps {
        echo '==== Deploy On Tomcat ====='
    }
}

```

```

    deploy(
        adapters: [
            tomcat9(
                credentialsId: 'tomcat-jenkins',
                url: 'http://172.31.33.74:8080'
            )
        ],
        contextPath: 'customerapp',
        war: "target/SimpleCustomerApp-${BUILD_NUMBER}-SNAPSHOT.war"
    )
}
}
}
}
}

```

10. Configure the Git repository:

Repository:

https://github.com/betawins/sabear_simplecutomerapp.git

Branch:

*/feature-1.1

11. The repository uses the **feature-1.1** branch.

12. Verify that the repository contains the required Maven project files such as:

pom.xml

src/

WebContent/

build.xml

Jenkinsfile

13. Configure **Maven** in Jenkins if required.

For this implementation, Maven was executed through the Docker image:

maven:3.9.9-eclipse-temurin-21

14. Configure **SonarQube** in:

Manage Jenkins

→ System

→ SonarQube servers

15. Configure the SonarQube server with the Jenkins server name:

SonarQube

16. The SonarQube server used for this implementation was:

http://172.31.12.251:9000/

17. Configure the SonarQube authentication token in Jenkins.

18. Configure **Nexus credentials** in Jenkins.

Credential ID:

nexus-admin

19. Configure Nexus using:

Nexus URL:

172.31.47.243:8081

20. The Nexus repository used was:

maven_releases

21. The Maven project values were:

Group ID:

com.javatpoint

Artifact ID:

SimpleCustomerApp

Packaging:

WAR

22. Configure the **Slack Notification** plugin and Slack integration in Jenkins.

23. Configure the **Tomcat credentials** in Jenkins.

Credential ID:

tomcat-jenkins

24. Configure Tomcat deployment using:

Tomcat URL:

http://172.31.33.74:8080

25. The pipeline performs the following stages:

1. Git Clone

↓

2. SonarQube Integration

↓

3. Maven Compilation

↓

4. Nexus Artifactory

↓

5. Slack Notification

↓

6. Deploy On Tomcat

26. The **Git Clone** stage checks out the feature-1.1 branch.

27. The **SonarQube Integration** stage performs code analysis using the Maven SonarQube plugin.

28. The SonarQube project was configured as:

Project Key:

customer-app

Project Name:

Customer-App

29. The **Maven Compilation** stage builds the application using:

```
mvn clean package
```

30. Maven generates the WAR file under:

```
target/
```

31. The generated WAR follows this naming format:

```
SimpleCustomerApp-${BUILD_NUMBER}-SNAPSHOT.war
```

For example:

```
SimpleCustomerApp-7-SNAPSHOT.war
```

32. Verify the generated WAR using:

```
ls -lh target/
```

33. The **Nexus Artifactory** stage uploads the WAR to Nexus.

34. Nexus uses the following repository:

```
maven_releases
```

35. The Nexus artifact uses:

Group ID:

```
com.javatpoint
```

Artifact ID:

```
SimpleCustomerApp
```

Version:

```
${BUILD_NUMBER}
```

Type:

```
war
```

36. The version uploaded to the **RELEASE** repository is `${BUILD_NUMBER}` because the `maven_releases` repository does not accept SNAPSHOT versions.

37. For example, Build #7 uploaded the artifact as:

SimpleCustomerApp-7.war

under:

com/javatpoint/SimpleCustomerApp/7/

38. The **Slack Notification** stage sends a success message after the Nexus upload.

39. Verify that the Slack notification is received successfully.

40. The **Deploy On Tomcat** stage deploys:

target/SimpleCustomerApp-\${BUILD_NUMBER}-SNAPSHOT.war

41. The Tomcat context path is:

customerapp

42. Therefore, the deployed application URL follows this format:

http://<Tomcat-Public-IP>:8080/customerapp/

43. Click **Save**.

44. Click **Build Now**.

45. Open the running build.

46. Click **Console Output**.

47. Verify that the **Git Clone** stage completed successfully.

48. Verify the Git branch:

feature-1.1

49. Verify the latest Git commit is displayed.

50. Verify that the **SonarQube Analysis** completed successfully.

51. Verify the SonarQube project:

customer-app

52. Open the SonarQube dashboard and verify the analysis results.

53. Verify the **Maven Compilation** stage.

54. Confirm the console shows:

BUILD SUCCESS

55. Verify the generated WAR inside:

target/

56. Verify the **Nexus Artifact Upload** stage.

57. Confirm the artifact was uploaded successfully to:

maven_releases

58. Verify the artifact in Nexus under:

com/javatpoint/SimpleCustomerApp/<BUILD_NUMBER>/

59. Verify the **Slack Notification** stage.

60. Confirm that the Slack success notification was received.

61. Verify the **Deploy On Tomcat** stage.

62. Confirm that the WAR deployment completed successfully.

63. Open the deployed application using:

http://<Tomcat-Public-IP>:8080/customerapp/

64. Confirm that the application is accessible from the browser.

65. Confirm the Jenkins build result:

Finished: SUCCESS



The screenshot shows the Jenkins web interface with the console output of a build. The browser address bar shows the URL: 100.54.71.26:8080/job/CustomerApp-Declarative-Pipeline/1/console. The Jenkins logo and job name 'CustomerApp-Declarative...' are visible at the top. The console output displays a series of file permissions and sizes for various files, followed by a 'Git Commit' message and a 'git log' command. The build concludes with the message 'Finished: SUCCESS'.

```
drwxr-xr-x, 6 jenkins jenkins 192 Oct 7 12:54 .
drwxr-xr-x, 7 jenkins jenkins 191 Oct 7 12:54 ..
drwxr-xr-x, 7 jenkins jenkins 146 Oct 7 12:54 .git
-rw-r--r--, 1 jenkins jenkins 4261 Oct 7 12:54 Jenkinsfile
-rw-r--r--, 1 jenkins jenkins 99 Oct 7 12:54 Service-API.dockerfile
drwxr-xr-x, 6 jenkins jenkins 174 Oct 7 12:54 WebContent
-rw-r--r--, 1 jenkins jenkins 4 Oct 7 12:54 XMLSave.txt
drwxr-xr-x, 6 jenkins jenkins 54 Oct 7 12:54 build
-rw-r--r--, 1 jenkins jenkins 23830 Oct 7 12:54 build.xml
-rw-r--r--, 1 jenkins jenkins 1050 Oct 7 12:54 pom.xml
-rw-r--r--, 1 jenkins jenkins 1003 Oct 7 12:54 sonar-project.properties
drwxr-xr-x, 3 jenkins jenkins 17 Oct 7 12:54 src
[Pipeline] echo
===== Git Commit =====
[Pipeline] sh
+ git log -1 --oneline
05222fe Update pom.xml
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

← → ↻ Not secure 100.54.71.26:8080/job/CustomerApp-Declarative-Pipeline/configure

Jenkins / CustomerApp-Declarative... / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script

Script ?

```
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('Git Clone') {
7       steps {
8         echo '==== Git Clone ====='
9
10        git branch: 'feature-1.1',
11        url: 'https://github.com/betawins/sabear_simplecustomerapp.git'
12
13        echo '==== Repository Files ====='
14        sh 'ls -la'
15      }
16    }
17  }
18}
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Save Apply

← → ↻ Not secure 100.54.71.26:8080/job/CustomerApp-Declarative-Pipeline/4/console

Jenkins / CustomerApp-Declarative... / #4

```
[INFO] SCM Publisher 1 source file to be analyzed
[INFO] SCM Publisher 1/1 source file have been analyzed (done) | time=251ms
[INFO] CPD Executor calculating CPD for 0 files
[INFO] CPD Executor CPD calculation finished (done) | time=0ms
[INFO] SCM revision ID '05222fef4b498f40dd73a8a1b6fbd0c619d398a'
[INFO] Analysis report generated in 221ms, dir size=293.2 kB
[INFO] Analysis report compressed in 16ms, zip size=31.6 kB
[INFO] Analysis report uploaded in 299ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://172.31.12.251:9000/dashboard?id=customer-app
[INFO] Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
[INFO] More about the report processing at http://172.31.12.251:9000/api/ce/task?id=95134486-35b2-478e-9476-453f9c00f421
[INFO] Analysis total time: 3.357 s
[INFO] SonarScanner Engine completed successfully
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 16.211 s
[INFO] Finished at: 2026-10-07T12:59:17Z
[INFO] -----
[Pipeline] }
[Pipeline] // withSonarQubeEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

← → ↻ Not secure 184.195.163.53:9000/projects

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

The way in which security, reliability, and maintainability counts and ratings are calculated has changed. [Learn more in SonarQube documentation](#)

SonarQube community Projects Issues Rules Quality profiles Quality gates Administration More

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Filters

Quality gate

Passed 3

Failed 0

Security

A ≥ 0 info issues 1

B ≥ 1 low issue 1

C ≥ 1 medium issue 1

D ≥ 1 high issue 0

E ≥ 1 blocker issue 0

Reliability

A ≥ 0 info issues 1

B ≥ 1 low issue 0

Search projects Perspective Overall Status Sort by Name 3 project(s)

Customer-App Public Passed

Last analysis: 10 seconds ago - 28 Lines of Code - XML

A 0 A 0 A 0 A — — 0.0%

Security Reliability Maintainability Hotspots Reviewed Coverage Duplications

Hiring-App Public Passed

Last analysis: 23 minutes ago - 45 Lines of Code - XML, Docker, ...

B 2 C 3 A 0 A — — 0.0%

Security Reliability Maintainability Hotspots Reviewed Coverage Duplications

VProfile-1 Public Passed

Last analysis: 1 day ago - 1.8k Lines of Code - JSP, Java, ...

Jenkins / CustomerApp-Declarative... / #5

Progress (1): 44 kB

Downloaded from central: <https://repo.maven.apache.org/maven2/asm/asm/3.3.1/asm-3.3.1.jar> (44 kB at 176 kB/s)

[INFO] Packaging webapp

[INFO] Assembling webapp [SimpleCustomerApp] in [/workspace/target/SimpleCustomerApp-5-SNAPSHOT]

[INFO] Processing war project

[INFO] Building war: /workspace/target/SimpleCustomerApp-5-SNAPSHOT.war

[INFO] BUILD SUCCESS

[INFO] Total time: 10.751 s

[INFO] Finished at: 2026-10-07T13:01:32Z

[INFO] -----

[Pipeline] echo

==== Build Artifacts =====

[Pipeline] sh

+ ls -lh target/

total 4.0K

drwxr-xr-x. 4 root root 37 Oct 7 13:01 SimpleCustomerApp-5-SNAPSHOT

-rw-r--r--. 1 root root 1.4K Oct 7 13:01 SimpleCustomerApp-5-SNAPSHOT.war

drwxr-xr-x. 2 root root 28 Oct 7 13:01 maven-archiver

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS

Jenkins / CustomerApp-Declarative... / #7

[Pipeline] stage

[Pipeline] { (Nexus Artifactory)

[Pipeline] echo

==== Nexus Artifact Upload =====

[Pipeline] nexusArtifactUploader

Uploading artifact SimpleCustomerApp-7-SNAPSHOT.war started....

GroupId: com.javatpoint

ArtifactId: SimpleCustomerApp

Classifier:

Type: war

Version: 7

File: SimpleCustomerApp-7-SNAPSHOT.war

Repository:maven_releases

Uploading: http://172.31.47.243:8081/repository/maven_releases/com/javatpoint/SimpleCustomerApp/7/SimpleCustomerApp-7.war

100 % completed (1.4 kB / 1.4 kB).

Uploaded: http://172.31.47.243:8081/repository/maven_releases/com/javatpoint/SimpleCustomerApp/7/SimpleCustomerApp-7.war (1.4 kB at 24 kB/s)

Uploading artifact SimpleCustomerApp-7-SNAPSHOT.war completed.

[Pipeline] }

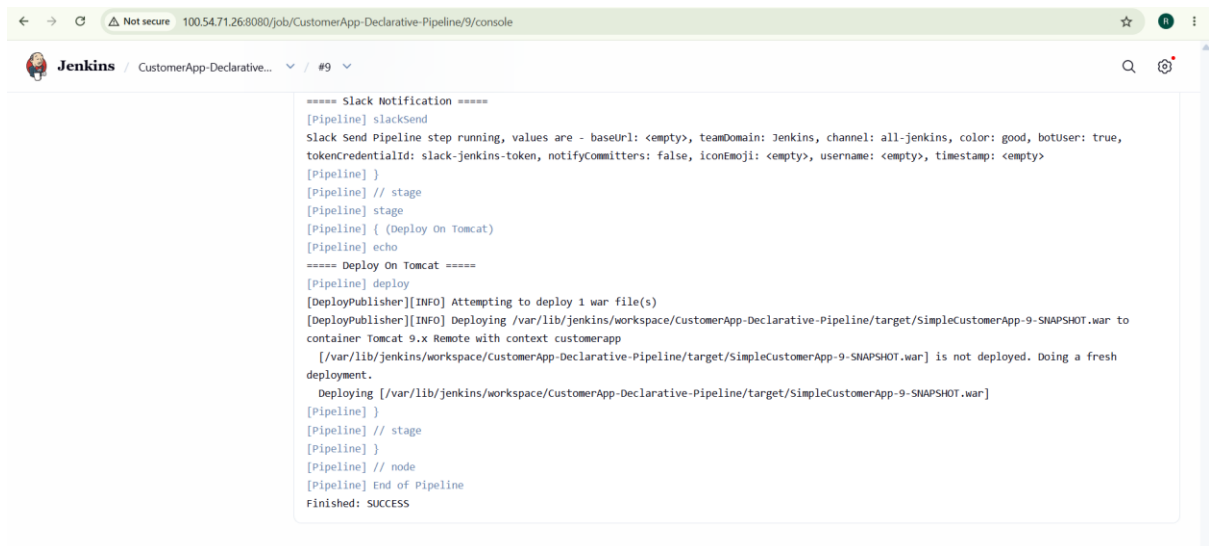
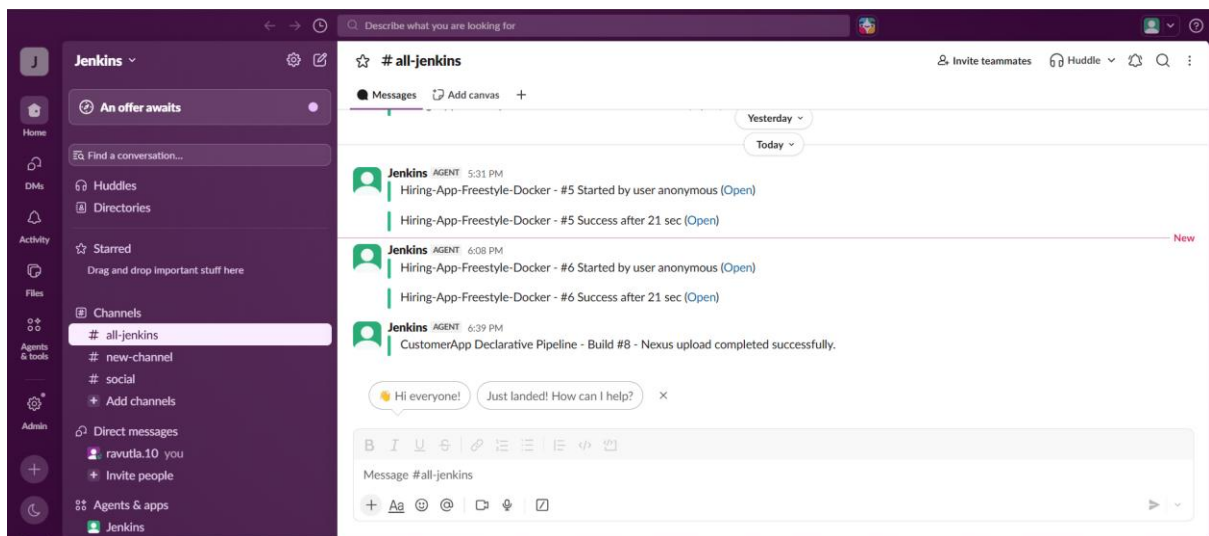
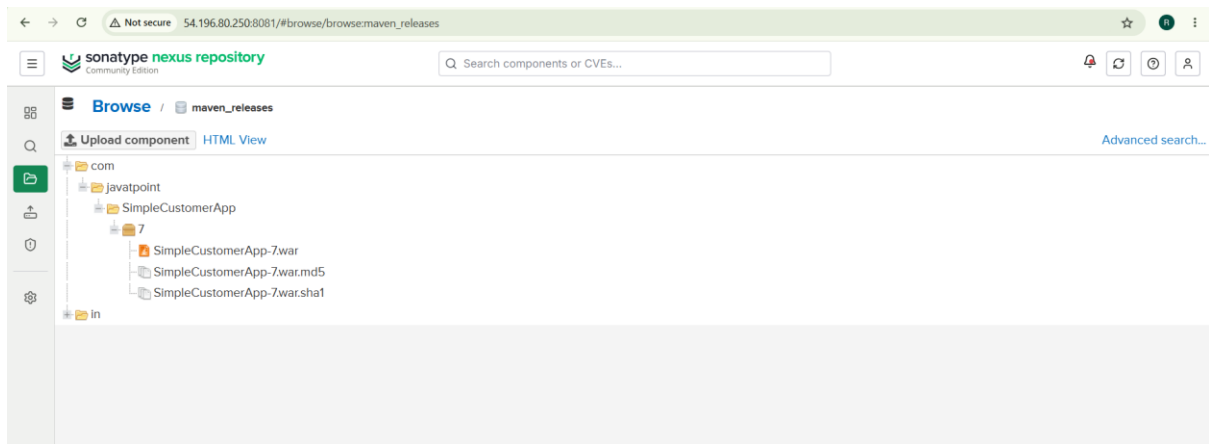
[Pipeline] // stage

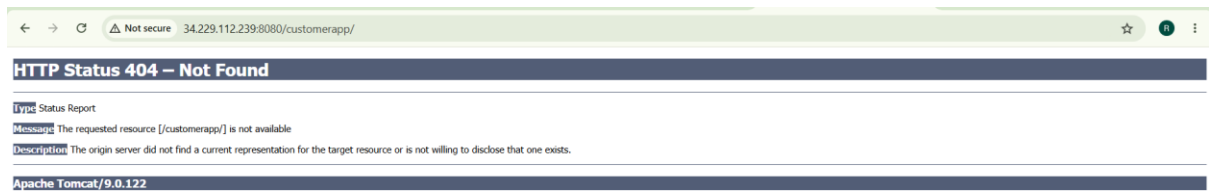
[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

Finished: SUCCESS





3) Setup a Jenkins CICD pipeline using Scripted pipeline using feature-1.1 branch.

Source Code:

https://github.com/betawins/sabear_simplecutomerapp/tree/feature-1.1

stages:

- **Git Clone**
- **SonarQube Integration**
- **Maven Compilation**
- **Nexus Artifactory**
- **Slack Notification**
- **Deploy On tomcat**

1. Create Jenkins Job

- Create a new **Pipeline** job.
- Job name: CustomerApp-Scripted-Pipeline
- Select **Pipeline script**.

2. Git Clone

- Clone the application repository from GitHub.
- Use branch: feature-1.1.
- Clean the workspace before cloning.

Process: GitHub → Jenkins

3. SonarQube Analysis

- Connect Jenkins with SonarQube.
- Run code analysis using Maven inside Docker.

Process: Source Code → SonarQube → Code Analysis

4. Maven Compilation

- Run Maven inside Docker.
- Execute clean package.
- Generate the WAR file.

Process: Source Code → Maven → WAR

5. Nexus Artifactory

- Upload the generated WAR file to Nexus.
- Repository: maven_releases
- Use Jenkins build number as the release version.

Process: WAR → Nexus

6. Slack Notification

- Send a Slack notification after successful Nexus upload.
- Include the pipeline name and build number.

Process: Nexus Success → Slack

7. Deploy on Tomcat

- Deploy the WAR file to Tomcat.
- Use the configured Tomcat credentials.
- Context path: customerapp-scripted.

Process: WAR → Tomcat → Application

8. Final Verification

- Verify all stages complete successfully.
- Jenkins should show:

Finished: SUCCESS

← → ↻ Not secure 44.215.103.121:8080/job/CustomerApp-Scripted-Pipeline/configure

Jenkins / CustomerApp-Scripted-Pi... / Configure

Configure

- General
- Triggers
- Pipeline**
- Advanced

Definition

Pipeline script

Script ?

1 try sample Pipeline...

☒ Use Groovy Sandbox ?

Pipeline Syntax

Save Apply

← → ↻ us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0ae7e7950f51101f&osUser=ec2-user&sshPort=22&address=...

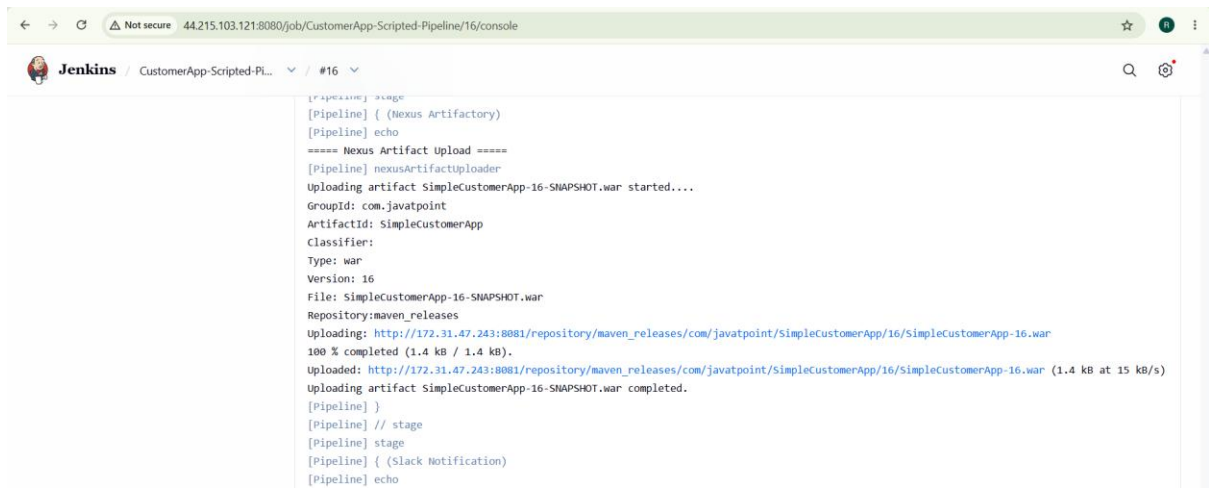
```
gging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.h
andler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener.UMASK=0027
-Dignore.endorsed.dirs=-classpath /opt/apache-tomcat-9.0.122/bin/bootstrap.jar:/opt/apache-tomcat-9.0.122/bin/tomcat-juli.jar -Dcat
alina.base=/opt/apache-tomcat-9.0.122 -Dcatalina.home=/opt/apache-tomcat-9.0.122 -Djava.io.tmpdir=/opt/apache-tomcat-9.0.122/temp org
.apache.catalina.startup.Bootstrap start
ec2-user 2681 2248 0 13:32 pts/0 00:00:00 grep --color=auto tomcat
[ec2-user@ip-172-31-33-74 ~]$ sudo ss -lntp | grep 8080
LISTEN 0 100 *:8080 *: * users: (("java",pid=2595,fd=45))
[ec2-user@ip-172-31-33-74 ~]$ curl -I http://localhost:8080
HTTP/1.1 200
Content-Type: text/html;charset=UTF-8
Date: Wed, 07 Oct 2026 13:32:52 GMT

[ec2-user@ip-172-31-33-74 ~]$ curl -4 -u jenkins http://127.0.0.1:8080/manager/text/list
Enter host password for user 'jenkins':
OK - Listed applications for virtual host [localhost]
/:running:0:ROOT
/examples:running:0:examples
/host-manager:running:0:host-manager
/hiring:running:0:hiring
/customerapp:running:0:customerapp
/manager:running:0:manager
/docs:running:0:docs
[ec2-user@ip-172-31-33-74 ~]$
```

Jenkins / CustomerApp-Scripted-Pi... / #16

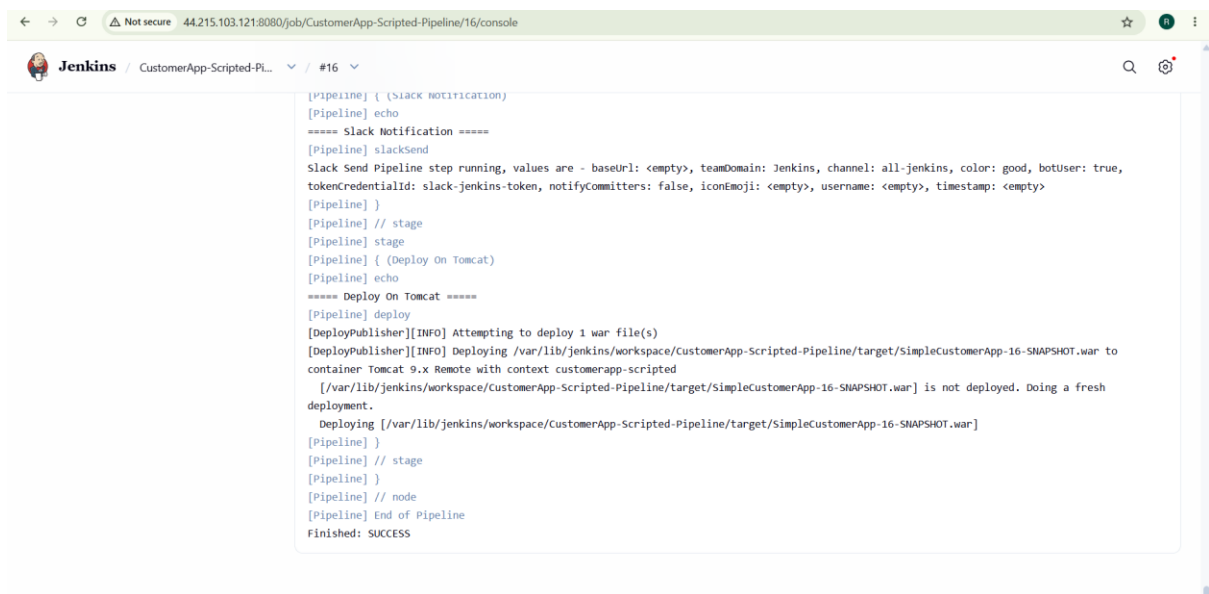
Downloaded from central: <https://repo.maven.apache.org/maven2/com/google/guava/guava/10.0.1/guava-10.0.1.jar> (1.5 MB at 4.1 MB/s)

```
[INFO] Packaging webapp
[INFO] Assembling webapp [SimpleCustomerApp] in [/workspace/target/SimpleCustomerApp-16-SNAPSHOT]
[INFO] Processing war project
[INFO] Building war: /workspace/target/SimpleCustomerApp-16-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 9.641 s
[INFO] Finished at: 2026-10-07T13:47:57Z
[INFO] -----
[Pipeline] echo
==== Build Artifacts ====
[Pipeline] sh
+ ls -lh target/
total 4.0K
drwxr-xr-x. 4 jenkins jenkins 37 Oct 7 13:47 SimpleCustomerApp-16-SNAPSHOT
-rw-r--r--. 1 jenkins jenkins 1.4K Oct 7 13:47 SimpleCustomerApp-16-SNAPSHOT.war
drwxr-xr-x. 2 jenkins jenkins 28 Oct 7 13:47 maven-archiver
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Nexus Artifactory)
[Pipeline] echo
```



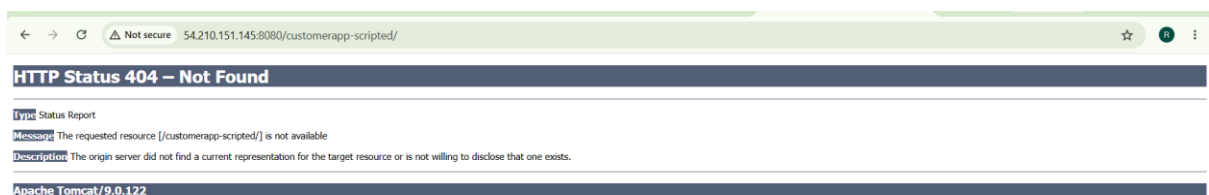
The screenshot shows the Jenkins console output for a pipeline stage. The output is as follows:

```
[Pipeline] { (Nexus Artifact)  
[Pipeline] echo  
==== Nexus Artifact Upload ====  
[Pipeline] nexusArtifactUploader  
Uploading artifact SimpleCustomerApp-16-SNAPSHOT.war started....  
GroupId: com.javatpoint  
ArtifactId: SimpleCustomerApp  
Classifier:  
Type: war  
Version: 16  
File: SimpleCustomerApp-16-SNAPSHOT.war  
Repository:maven_releases  
Uploading: http://172.31.47.243:8081/repository/maven_releases/com/javatpoint/SimpleCustomerApp/16/SimpleCustomerApp-16.war  
100 % completed (1.4 kB / 1.4 kB).  
Uploaded: http://172.31.47.243:8081/repository/maven_releases/com/javatpoint/SimpleCustomerApp/16/SimpleCustomerApp-16.war (1.4 kB at 15 kB/s)  
Uploading artifact SimpleCustomerApp-16-SNAPSHOT.war completed.  
[Pipeline] }  
[Pipeline] // stage  
[Pipeline] stage  
[Pipeline] { (Slack Notification)  
[Pipeline] echo
```



The screenshot shows the Jenkins console output for a pipeline stage. The output is as follows:

```
[Pipeline] { (Slack Notification)  
[Pipeline] echo  
==== Slack Notification ====  
[Pipeline] slackSend  
Slack Send Pipeline step running, values are - baseUrl: <empty>, teamDomain: Jenkins, channel: all-jenkins, color: good, botUser: true,  
tokenCredentialId: slack-jenkins-token, notifyCommitters: false, iconEmoji: <empty>, username: <empty>, timestamp: <empty>  
[Pipeline] }  
[Pipeline] // stage  
[Pipeline] stage  
[Pipeline] { (Deploy On Tomcat)  
[Pipeline] echo  
==== Deploy On Tomcat ====  
[Pipeline] deploy  
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)  
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/CustomerApp-Scripted-Pipeline/target/SimpleCustomerApp-16-SNAPSHOT.war to  
container Tomcat 9.x Remote with context customerapp-scripted  
[/var/lib/jenkins/workspace/CustomerApp-Scripted-Pipeline/target/SimpleCustomerApp-16-SNAPSHOT.war] is not deployed. Doing a fresh  
deployment.  
Deploying [/var/lib/jenkins/workspace/CustomerApp-Scripted-Pipeline/target/SimpleCustomerApp-16-SNAPSHOT.war]  
[Pipeline] }  
[Pipeline] // stage  
[Pipeline] }  
[Pipeline] // node  
[Pipeline] End of Pipeline  
Finished: SUCCESS
```



4) Write sample skeleton of pipelines.

Git Clone → SonarQube → Maven → Nexus → Slack → Tomcat

1. Declarative Pipeline Skeleton

```
pipeline {  
  
    agent any
```

```
stages {

    stage('Git Clone') {

        steps {

            // Clone source code

        }

    }

    stage('SonarQube Analysis') {

        steps {

            // Run SonarQube analysis

        }

    }

    stage('Maven Compilation') {

        steps {

            // Run Maven build

        }

    }

    stage('Nexus Artifactory') {

        steps {

            // Upload artifact to Nexus

        }

    }

    stage('Slack Notification') {

        steps {
```

```
        // Send Slack notification
    }
}

stage('Deploy on Tomcat') {
    steps {
        // Deploy application to Tomcat
    }
}
}
```

2. Scripted Pipeline Skeleton

```
node {

    stage('Git Clone') {
        // Clone source code
    }

    stage('SonarQube Analysis') {
        // Run SonarQube analysis
    }

    stage('Maven Compilation') {
        // Run Maven build
    }

    stage('Nexus Artifactory') {
        // Upload artifact to Nexus
    }
}
```

```

    }

    stage('Slack Notification') {
        // Send Slack notification
    }

    stage('Deploy on Tomcat') {
        // Deploy application to Tomcat
    }
}

```

Key Difference

Declarative	Scripted
Uses pipeline {}	Uses node {}
Structured format	More flexible
Easier to read	More programming-oriented
Has stages and steps blocks	Stages are written directly

5) Create a parameterized job in Jenkins.

Source Code: <https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git>

Implementation Steps:

1. Create Jenkins Job

- Create a **Freestyle Project**.
- Job name: Parameterized-Jenkins-Job.

2. Configure Parameter

- Enable **This project is parameterized**.
- Add a **String Parameter**.
- Parameter name: BRANCH
- Default value: master.
- This parameter allows the user to select which Git branch to build.

3. Configure Git Repository

- Select **Git** under Source Code Management.
- Repository:
`https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git`
- Branch Specifier:
`*/${BRANCH}`
- No credentials are required because the repository is public.

4. Add Build Step

- Select **Execute shell**.
- Use the parameter to display the selected branch and workspace.

Process:

BRANCH Parameter → Git Checkout → Build Step

5. Test with master

- Select **Build with Parameters**.
- Set:

BRANCH = master

- Start the build.
- Jenkins successfully checks out the master branch.

6. Test with Another Branch

- Select **Build with Parameters** again.
- Set:

BRANCH = pom.xmlversion-testing

- Start the build.
- Jenkins successfully checks out the selected branch.

7. Final Verification

Both branch selections completed successfully.

Overall Process:

User selects Branch → Jenkins Parameter → Git Checkout → Build Execution → SUCCESS

The screenshot shows the Jenkins configuration page for a job named 'Parameterized-Jenkins-Job'. The 'Configure' tab is active, and the 'General' sub-tab is selected. The 'This project is parameterised' checkbox is checked. A 'String Parameter' is configured with the name 'BRANCH', a default value of 'master', and a description 'Git branch to build'. The 'Trim the string' checkbox is unchecked. The 'Save' button is highlighted in blue.

The screenshot shows the Jenkins configuration page for the same job, with the 'Source Code Management' sub-tab selected. The 'Git' option is chosen. The 'Repository URL' is set to 'https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git'. The 'Credentials' dropdown is set to 'none'. The 'Advanced' section is expanded. The 'Branches to build' section is visible, with the 'Branch Specifier (blank for \'any\')' set to '*/\$(BRANCH)'. The 'Save' button is highlighted in blue.

← → ↻ Not secure 3.238.9.45:8080/job/Parameterized-Jenkins-Job/configure

Jenkins / Parameterized-Jenkins-Job / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See the list of available environment variables

```
echo "Building branch: $BRANCH"
echo "Workspace: $WORKSPACE"
git branch --show-current
```

Advanced ▾

+ Add build step

Save Apply

aws Search [Alt+S] United States (N. Virginia) Poojith R (956035085469) Poojith R

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Thu Oct  8 06:12:50 2026 from 18.206.107.27
[ec2-user@ip-172-31-11-114 ~]$ git ls-remote --heads https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git
fd245ec10e1d712908021555cb7b3befe4c37479      refs/heads/master
1f1a65e35dc813f3e0164d4e14c895b706da869c      refs/heads/pom.xmlversion-testing
[ec2-user@ip-172-31-11-114 ~]$
```

← → × Not secure 3.238.9.45:8080/job/Parameterized-Jenkins-Job/build?delay=0sec

Jenkins / Parameterized-Jenkins-Job

Status

Changes

Workspace

Build with Parameters

Delete Project

Configure

Rename

Project Parameterized-Jenkins-Job

This build requires parameters:

BRANCH

Git branch to build

master

Build Cancel

Builds > Filter Loading builds...

← → ↻ ⚠ Not secure 3.238.9.45:8080/job/Parameterized-Jenkins-Job/2/console

Jenkins / Parameterized-Jenkins-Job / #2

Console Output

- Delete build '#2'
- Parameters
- Git Build Data
- Edit build information
- Previous Build

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Parameterized-Jenkins-Job
The recommended git tool is: NONE
No credentials specified
> /usr/bin/git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Parameterized-Jenkins-Job/.git # timeout=10
Fetching changes from the remote Git repository
> /usr/bin/git config remote.origin.url https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git # timeout=10
Fetching upstream changes from https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git
> /usr/bin/git --version # timeout=10
> git --version # 'git version 2.50.1'
> /usr/bin/git fetch --tags --force --progress -- https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> /usr/bin/git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision fd245ec18e1d712908821555cb7b3bfe4c37479 (refs/remotes/origin/master)
> /usr/bin/git config core.sparsecheckout # timeout=10
> /usr/bin/git checkout -f fd245ec18e1d712908821555cb7b3bfe4c37479 # timeout=10
Commit message: "Update Jenkinsfile"
First time build. Skipping changelog.
[Parameterized-Jenkins-Job] $ /bin/sh -xe /tmp/jenkins1707015666008959466.sh
+ echo 'Building branch: master'
Building branch: master
+ echo 'workspace: /var/lib/jenkins/workspace/Parameterized-Jenkins-Job'
workspace: /var/lib/jenkins/workspace/Parameterized-Jenkins-Job
+ git branch --show-current
Finished: SUCCESS
```

← → ↻ ⚠ Not secure 3.238.9.45:8080/job/Parameterized-Jenkins-Job/build?delay=0sec

Jenkins / Parameterized-Jenkins-Job

Status

- Changes
- Workspace
- Build with Parameters
- Delete Project
- Configure
- Rename

Project Parameterized-Jenkins-Job

This build requires parameters:

BRANCH

Git branch to build

pom.xmlversion-testing

▶ Build Cancel

Builds > Filter /

Today

- #3 06:40
- #2 06:38
- #1 06:34

← → ↻ ⚠ Not secure 3.238.9.45:8080/job/Parameterized-Jenkins-Job/3/console

Jenkins / Parameterized-Jenkins-Job / #3

Changes

Console Output

- Delete build '#3'
- Parameters
- Git Build Data
- Edit build information
- Previous Build

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Parameterized-Jenkins-Job
The recommended git tool is: NONE
No credentials specified
> /usr/bin/git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Parameterized-Jenkins-Job/.git # timeout=10
Fetching changes from the remote Git repository
> /usr/bin/git config remote.origin.url https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git # timeout=10
Fetching upstream changes from https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git
> /usr/bin/git --version # timeout=10
> git --version # 'git version 2.50.1'
> /usr/bin/git fetch --tags --force --progress -- https://github.com/betawins/spring3-mvc-maven-xml-hello-world-1.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> /usr/bin/git rev-parse refs/remotes/origin/pom.xmlversion-testing^{commit} # timeout=10
Checking out Revision 1f1a65e35dc813f3e0164d4e14c895b706da869c (refs/remotes/origin/pom.xmlversion-testing)
> /usr/bin/git config core.sparsecheckout # timeout=10
> /usr/bin/git checkout -f 1f1a65e35dc813f3e0164d4e14c895b706da869c # timeout=10
Commit message: "updating pom.xml file"
First time build. Skipping changelog.
[Parameterized-Jenkins-Job] $ /bin/sh -xe /tmp/jenkins10767316005264768193.sh
+ echo 'Building branch: pom.xmlversion-testing'
Building branch: pom.xmlversion-testing
+ echo 'workspace: /var/lib/jenkins/workspace/Parameterized-Jenkins-Job'
workspace: /var/lib/jenkins/workspace/Parameterized-Jenkins-Job
+ git branch --show-current
Finished: SUCCESS
```

6) Setup one slave machine for Jenkins.

Jenkins Official Documentation

Implementation Steps:

1. Create Jenkins Agent EC2 Instance

- Launch a separate EC2 instance for the Jenkins Agent.
- Install **Java 21**.
- Create a dedicated jenkins user.

2. Create Jenkins User on Agent

```
sudo useradd -m -s /bin/bash jenkins
```

Verify:

```
id jenkins
```

3. Enable SSH on Agent

```
sudo systemctl enable --now sshd
```

Prepare the SSH directory:

```
sudo mkdir -p /home/jenkins/.ssh
```

```
sudo chown -R jenkins:jenkins /home/jenkins/.ssh
```

```
sudo chmod 700 /home/jenkins/.ssh
```

4. Generate SSH Key on Jenkins Controller

Generate an SSH key for the Jenkins user:

```
sudo -u jenkins ssh-keygen -t ed25519
```

The key is stored under the Jenkins user's .ssh directory.

5. Configure SSH Key Authentication

Copy the public key from the Jenkins Controller to the Agent:

```
sudo -u jenkins ssh-copy-id jenkins@<Private IP of Agent>
```

Test passwordless SSH:

```
sudo -u jenkins ssh jenkins@<Private IP of Agent>
```

Verify the Agent hostname:

hostname

Exit:

exit

6. Create Agent Node in Jenkins

Go to:

Jenkins → Manage Jenkins → Nodes → New Node

Configure:

- **Node name:** jenkins-agent
- **Type:** Permanent Agent
- **Executors:** 2
- **Remote root directory:** /home/jenkins
- **Labels:** linux-agent
- **Usage:** Only build jobs with label expressions matching this node
- **Launch method:** Launch agents via SSH
- **Host:** <Private IP of Agent>
- **Username:** jenkins
- **Private key:** Jenkins Controller SSH private key
- **Host Key Verification:** Known hosts file

Save the configuration.

7. Resolve Disk Space Warning

The Agent's /tmp filesystem was only **457 MB**, while Jenkins' default threshold was **1 GiB**.

The Jenkins disk-space monitoring threshold was changed from:

1 GiB → 100 MiB

The warning then disappeared.

8. Test Jenkins Agent

Create a Freestyle job:

Job name: Agent-Test

Configure:

Restrict where this project can be run:

linux-agent

Add **Execute shell**:

echo "Running on Jenkins Agent"

hostname

whoami

java -version

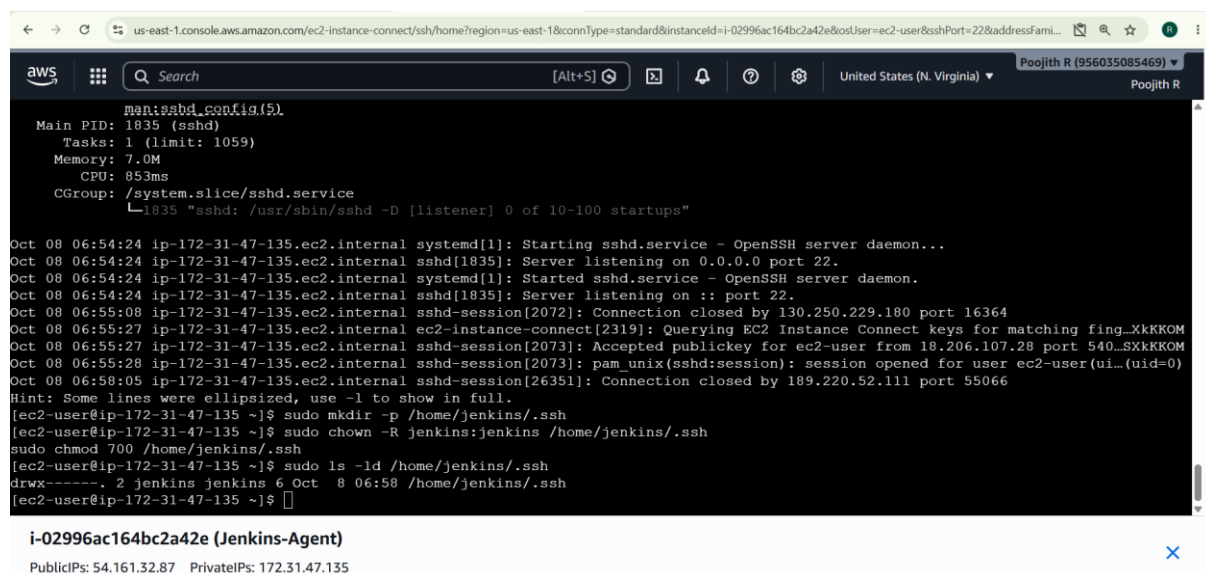
Run the job.

9. Verify Result

The job successfully executed on the Jenkins Agent and returned:

Finished: SUCCESS

The hostname confirmed that the job was executed on the Agent rather than the Jenkins Controller.



```
man:sshd.config(5)
Main PID: 1035 (sshd)
Tasks: 1 (limit: 1059)
Memory: 7.0M
CPU: 853ms
CGroup: /system.slice/sshd.service
└─1035 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 08 06:54:24 ip-172-31-47-135.ec2.internal systemd[1]: Starting sshd.service - OpenSSH server daemon...
Oct 08 06:54:24 ip-172-31-47-135.ec2.internal sshd[1035]: Server listening on 0.0.0.0 port 22.
Oct 08 06:54:24 ip-172-31-47-135.ec2.internal systemd[1]: Started sshd.service - OpenSSH server daemon.
Oct 08 06:54:24 ip-172-31-47-135.ec2.internal sshd[1035]: Server listening on :: port 22.
Oct 08 06:55:08 ip-172-31-47-135.ec2.internal sshd-session[2072]: Connection closed by 130.250.229.180 port 16364
Oct 08 06:55:27 ip-172-31-47-135.ec2.internal ec2-instance-connect[2319]: Querying EC2 Instance Connect keys for matching fing_XkKKOM
Oct 08 06:55:27 ip-172-31-47-135.ec2.internal sshd-session[2073]: Accepted pubkey for ec2-user from 18.206.107.28 port 540..SXkKKOM
Oct 08 06:55:28 ip-172-31-47-135.ec2.internal sshd-session[2073]: pam_unix(sshd:session): session opened for user ec2-user(ui...(uid=0)
Oct 08 06:58:05 ip-172-31-47-135.ec2.internal sshd-session[26351]: Connection closed by 189.220.52.111 port 55066
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@ip-172-31-47-135 ~]$ sudo mkdir -p /home/jenkins/.ssh
[ec2-user@ip-172-31-47-135 ~]$ sudo chown -R jenkins:jenkins /home/jenkins/.ssh
sudo chmod 700 /home/jenkins/.ssh
[ec2-user@ip-172-31-47-135 ~]$ sudo ls -ld /home/jenkins/.ssh
drwx-----. 2 jenkins jenkins 6 Oct 8 06:58 /home/jenkins/.ssh
[ec2-user@ip-172-31-47-135 ~]$
```

i-02996ac164bc2a42e (Jenkins-Agent)

PublicIPs: 54.161.32.87 PrivateIPs: 172.31.47.135

← → ↻ ⚠ Not secure 3.238.9.45:8080/manage/computer/createItem

Jenkins

Manage Jenkins

Nodes

🔍 ⚙

Jenkins build agent

Plain text Preview

Number of executors ?
2

Remote root directory ?
/home/jenkins

Labels ?
linux-agent

Usage ?
Only build jobs with label expressions matching this node

Launch method ?
Launch agents via SSH

Save

← → ↻ ⚠ Not secure 3.238.9.45:8080/manage/computer/createItem

Jenkins

Manage Jenkins

Nodes

🔍 ⚙

Launch method ?
Launch agents via SSH

Host ?
172.31.47.135

Credentials ?
jenkins (Jenkins-slave) + Add

Host Key Verification Strategy ?
Known hosts file Verification Strategy

Advanced

Availability ?
Keep this agent online as much as possible

Save

← → ↻ ⚠ Not secure 3.238.9.45:8080/manage/computer/

Jenkins

Manage Jenkins

Nodes

🔍 ⚙

New Node

Configure Monitors

↻

Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
🖥	Built-In Node	Linux (amd64)	In sync	26.04 GiB	0 B	1.86 GiB	0ms
🖥	jenkins-agent		N/A	N/A	N/A	N/A	N/A
last checked		6 ms	5 ms	5 ms	4 ms	4 ms	59 min

Icon: S M L

Legend

 Jenkins

Manage Jenkins

Nodes

jenkins-agent

Search

Settings

Status

Delete Agent

Configure

Build History

Load Statistics

Script Console

Log

System Information

Disconnect

Agent jenkins-agent

Jenkins build agent

Monitoring Data

Labels

linux-agent

Projects tied to jenkins-agent

None

Edit description

Mark this node temporarily offline

Help

Build Executor Status

0/2

 Jenkins

Agent-Test

Configure

Search

Settings

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Discard old builds

☐ GitHub project

☐ This project is parameterised

☐ Throttle builds

☐ Execute concurrent builds if necessary

☒ Restrict where this project can be run

Label Expression

linux-agent

Label linux-agent matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☒ None

☐ Git

Save

Apply

 Jenkins

Agent-Test

Configure

Search

Settings

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell

Command

See the list of available environment variables

echo "Running on Jenkins Agent"

hostname

whoami

java -version

Advanced

Save

Apply

